



$$V_{OC} = 400V. \quad I_{OC} = \dots\dots\dots A, \quad W_{OC} = \dots\dots\dots W$$

$$V_{SC} = \dots\dots\dots V. \quad I_{SC} = \dots\dots\dots V. \quad W_{SC} = \dots\dots\dots W$$

Per phase values are

$$V_0 = V_{OC} = \dots\dots\dots V \quad I_0 = \frac{I_{OC}}{\sqrt{3}} = \dots\dots\dots A$$

$$V_S = V_{SC} = \dots\dots\dots V \quad I_S = \frac{I_{SC}}{\sqrt{3}} = \dots\dots\dots A$$

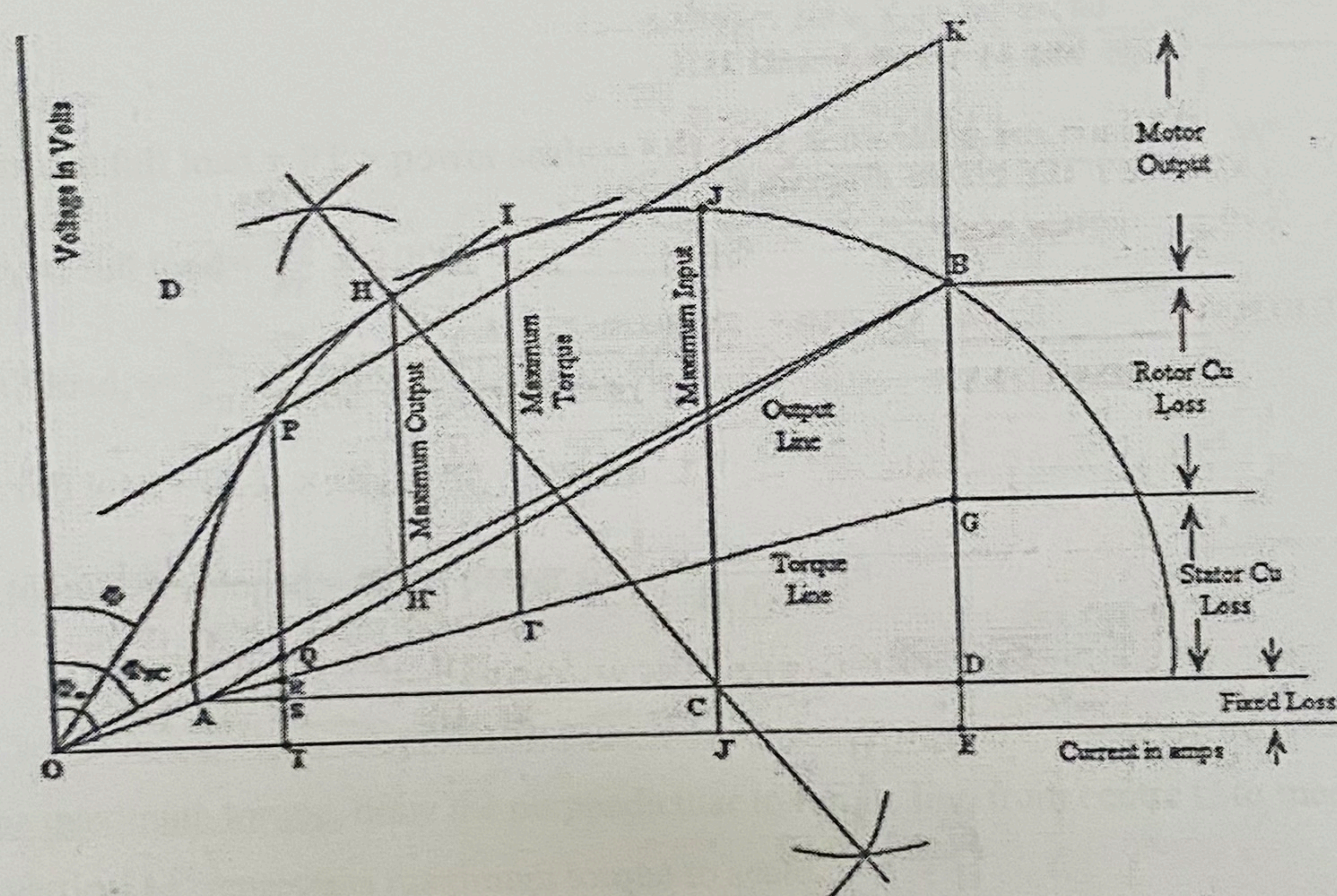
$$R_{DC} = \dots\dots\dots \Omega$$

$$R_1 = \frac{3}{2} \times 1.2 \times R_{DC} = \dots\dots\dots \Omega$$

$$R_{01} = \frac{W_{SC}}{3I_S^2} = \dots\dots\dots \Omega$$

$$R'_2 = R_{01} - R_1 = \dots\dots\dots \Omega$$

$$\frac{BG}{BD} = \frac{R'_2}{R_{01}}; \quad BG = \dots\dots\dots \times BD$$



Selection of current and power scale

$$\text{Current scale} = 1\text{cm} = \dots\dots\dots A \quad (\text{Take the current as large as possible, } 1\text{cm} = 1 \text{ or } 1.5A)$$

$$I_0 = \dots\dots\dots A \quad (= \dots\dots\dots \text{cm})$$

$$I_{SN} = I_{SC} \left(\frac{V_{rated}}{V_{SC}} \right) = \dots\dots\dots A \quad (= \dots\dots\dots \text{cm})$$

$$\phi_0 = \cos^{-1} \left(\frac{W_{OC}}{\sqrt{3}V_0 I_0} \right) = \dots\dots\dots$$

$$\phi_{SC} = \cos^{-1} \left(\frac{W_{SC}}{\sqrt{3}V_{SC} I_{SC}} \right) = \dots\dots\dots$$